

The Four Loops

A field guide to the loop architectures running AI agent systems in 2026, and the one ring above them that still needs a human.

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01

One Word, Four Systems

In June 2026, "loop" became AI engineering's word of the month. Peter Steinberger said stop prompting coding agents, start designing loops that prompt them. Boris Cherny of Anthropic said it on stage: "I write loops, the loops do the work." Addy Osmani, swyx, and LangChain each published essays on it within a ten day span, and the AI Engineer World's Fair closed with an hour-long debate over whether the hype has outrun what actually works.

The problem: none of them are describing the same thing. At least four distinct architectures are hiding behind that one word, plus a fifth ring above all of them that nobody had bothered to name.

A loop without a feedback signal isn't a loop, it's a pipeline. Fan-out patterns like Cognition's Agentic MapReduce dispatch, gather, and validate, but nothing feeds back into a next cycle. That's a topology you deploy inside a loop, not a loop of its own.

This guide names all five rings: what each one iterates on, what closes it, and where the real disagreement at AIEWF actually sat, which was never about the loops themselves, but about who gets to run the top one.

02

The Four Loop Architectures

LOOP 01

Execution Loop

The agent's own act, observe, decide cycle inside a single task. The innermost loop, and the only one most people picture when they say "agent."

- Iterates on: steps within one task
- Closes on: environment feedback, or the agent's own say-so
- Risk: ends when the agent decides it's done, whether or not it actually is

LOOP 02

Task Loop (Ralph Loop)

Wipes the context and restarts the agent against the same spec, over and over, one fresh context window per run.

- Iterates on: a single artifact or spec
- Closes on: spec compliance and passing tests
- The waste is the point: it prevents context rot and silent compaction damage

LOOP 03

Product Loop

The "software factory." The agent works across an entire codebase and backlog, continuously, picking up what nobody has even ticketed yet.

- Iterates on: the codebase and its backlog
- Closes on: signals from outside the repo, issues, logs, review outcomes
- Human role: configurable, not fixed

LOOP 04

System Loop

"Autoresearch." An outer loop that studies and improves the primary system itself: its own prompts, harness, and evals.

- Iterates on: prompts, harness, model choice, the evals themselves
- Closes on: evals, judges, and an explicit ask-a-human signal
- Even Meta's flagship version still has final configs chosen by hand

03 + 04

The Oversight Loop & Setting The Dial

Above all four sits a fifth ring with no exit condition listed: set goals, allocate budget, cull work. Its verbs are yours. It's the one ring where a human should live, because agency, not capability, is what it holds.

Oversight Loop → sets goals, allocates budget, culls work. no exit condition.

System Loop → improves the loop itself

Product Loop → continuous work across the codebase

Task Loop → restart until the spec passes

Execution Loop → act, observe, repeat, inside one task

CAMP A

Turn The Dial Up

- Pick checkpoints deliberately, ratchet autonomy as trust accumulates (Zach Lloyd)
- Warp raised its own auto-merge rate from 20% toward 60%
- "Build orchestras before factories": keep a human conductor (Roland Gavrilescu)
- Anthropic's internal Tag now authors 65% of one product team's code

DIAL UP

CAMP B

The Dial Has A Stop

- Those who delegate understanding get replaced by the agent (Geoffrey Litt)
- "There is no auto, and there will be no auto" (Paul Bakaous)
- Step down an abstraction level, not up (Dex Horthy)
- Even Anthropic's own Tag team reports being bottlenecked on human review

DIAL STOP

The real question was never which camp wins. It's what signal tells you, on each of the five rings, when to move the dial.